

T1489 + T1490 Service Stop and Inhibit System Recovery

1. Executive Summary

Chuỗi kiểm thử gồm hai Atomic Test thuộc T1489 và năm Atomic Test thuộc T1490 được thực hiện trên máy Windows 10 victim.

LimaCharlie Timeline ghi nhận đầy đủ bảy chuỗi lệnh cốt lõi:

- Dừng dịch vụ bằng sc.exe.
- Kết thúc tiến trình dịch vụ bằng taskkill.exe.
- Xóa Shadow Copy bằng vssadmin.exe.
- Xóa Shadow Copy bằng WMIC.exe.
- Xóa Windows Backup Catalog bằng wbadmin.exe.
- Vô hiệu hóa System Restore qua Registry.
- Thay đổi Shadow Storage bằng vssadmin.exe.

Splunk nhận được 23 detection trong cửa sổ quan sát. Trong đó có bốn detection liên quan đến recovery inhibition và một detection liên quan đến service stop.

Kết luận: True Positive - Authorized Simulation. Chuỗi hành vi có mức độ nghiêm trọng cao vì phù hợp với giai đoạn chuẩn bị trước mã hóa dữ liệu của ransomware.

2. Incident Classification

Hạng mục	Kết quả
Case Name	Service Stop and Recovery Inhibition Activity
Classification	True Positive - Authorized Simulation
MITRE Tactic	Impact
MITRE Techniques	T1489 và T1490
Atomic Tests	T1489 #1, #3 và T1490 #1, #2, #3, #9, #10
Affected Host	DESKTOP-JI0TPU0
Executing User	DESKTOP-JI0TPU0\longday
Source IP	172.16.1.10
Execution Date	28/07/2026
Analyst Timeline Source	LimaCharlie Timeline
Recommended Severity	High
Confidence	High đối với command execution
System Impact	Not Verified
Business Impact	None
Containment	Not Required
Case Status	Close after evidence and cleanup validation

3. Test Objective and Scope

Mục tiêu kiểm thử:

- Mô phỏng hành vi dừng dịch vụ trước giai đoạn gây tác động.
- Mô phỏng nhiều phương thức làm suy giảm khả năng phục hồi của Windows.
- Đánh giá process visibility của LimaCharlie.
- Kiểm tra rule detection cho T1489 và T1490.
- Xác nhận log được chuyển sang Splunk.
- Xác định khoảng trống giữa quan sát lệnh và xác nhận tác động.
- Đánh giá khả năng tương quan nhiều hành vi Impact trên cùng endpoint.

Phạm vi kiểm thử chỉ giới hạn trên máy Windows 10 victim thuộc Super SOC/SOAR Home Lab.

4. Test Execution and Detection Summary

Bảy test đều để lại process telemetry. Detection coverage hiện tại được xác nhận cho năm trong bảy test.

```
PS C:\Windows\system32> Invoke-AtomicTest T1490 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1490-1 Windows - Delete Volume Shadow Copies
T1490-2 Windows - Delete Volume Shadow Copies via WMI
T1490-3 Windows - wbadm Delete Windows Backup Catalog
T1490-4 Windows - Disable Windows Recovery Console Repair
T1490-5 Windows - Delete Volume Shadow Copies via WMI with PowerShell
T1490-6 Windows - Delete Backup Files
T1490-7 Windows - wbadm Delete systemstatebackup
T1490-8 Windows - Disable the SR scheduled task
T1490-9 Disable System Restore Through Registry
T1490-10 Windows - vssadmin Resize Shadowstorage Volume
T1490-11 Modify VSS Service Permissions
T1490-13 Windows - Delete Volume Shadow Copies via Diskshadow
PS C:\Windows\system32> Invoke-AtomicTest T1489 -ShowDetailsBrief
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

T1489-1 Windows - Stop service using Service Controller
T1489-2 Windows - Stop service using net.exe
T1489-3 Windows - Stop service by killing process
PS C:\Windows\system32>
```

```
T1489-3 Windows - Stop service by killing process
PS C:\Windows\system32> Invoke-AtomicTest T1489 -TestNumbers 1,3
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1489-1 Windows - Stop service using Service Controller
SERVICE_NAME: spooler
        TYPE               : 110  WIN32_OWN_PROCESS (interactive)
        STATE                : 3    STOP_PENDING
                           (NOT_STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0    (0x0)
        SERVICE_EXIT_CODE   : 0    (0x0)
        CHECKPOINT           : 0x3
        WAIT_HINT            : 0x4e20
Exit code: 0
Done executing test: T1489-1 Windows - Stop service using Service Controller
Executing test: T1489-3 Windows - Stop service by killing process
ERROR: The process "spoolsv.exe" not found.
Exit code: 128
Done executing test: T1489-3 Windows - Stop service by killing process
PS C:\Windows\system32> Invoke-AtomicTest T1490 -TestNumbers 1,2,9,10,3
PathToAtomicsFolder = C:\AtomicRedTeam\atomics

Executing test: T1490-1 Windows - Delete Volume Shadow Copies
vssadmin 1.1 - Volume Shadow Copy Service administrative command-line tool
(C) Copyright 2001-2013 Microsoft Corp.
No items found that satisfy the query.
Exit code: 1
Done executing test: T1490-1 Windows - Delete Volume Shadow Copies
Executing test: T1490-2 Windows - Delete Volume Shadow Copies via WMI
No Instance(s) Available.
Exit code: 0
Done executing test: T1490-2 Windows - Delete Volume Shadow Copies via WMI
Executing test: T1490-3 Windows - wbadm Delete Windows Backup Catalog
wbadmin 1.0 - Backup command-line tool
(C) Copyright Microsoft Corporation. All rights reserved.
The backup catalog has been successfully deleted.
Exit code: 0
Done executing test: T1490-3 Windows - wbadm Delete Windows Backup Catalog
Executing test: T1490-9 Disable System Restore Through Registry
The operation completed successfully.
The operation completed successfully.
The operation completed successfully.
The operation completed successfully.
Exit code: 0
Done executing test: T1490-9 Disable System Restore Through Registry
Executing test: T1490-10 Windows - vssadmin Resize Shadowstorage Volume
vssadmin 1.1 - Volume Shadow Copy Service administrative command-line tool
(C) Copyright 2001-2013 Microsoft Corp.
Successfully resized the shadow copy storage association
Exit code: 0
Done executing test: T1490-10 Windows - vssadmin Resize Shadowstorage Volume
PS C:\Windows\system32>
```

LimaCharlie EDR - SOC Operations Center

Real-time EDR monitoring dashboard for LimaCharlie detections with MITRE ATT&CK mapping, endpoint analysis, and suspicious process tracking.

Time Ran...

Last 15 minutes

Total Detections

23

Unique Endpoints

1

High/Critical Alerts

4

MITRE Techniques

7

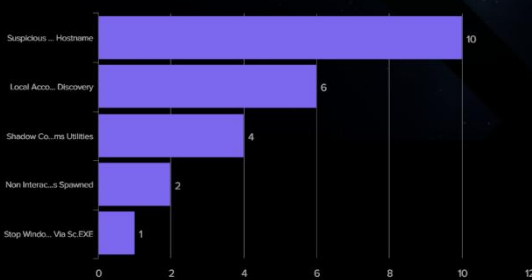
Active D&R Rules

5

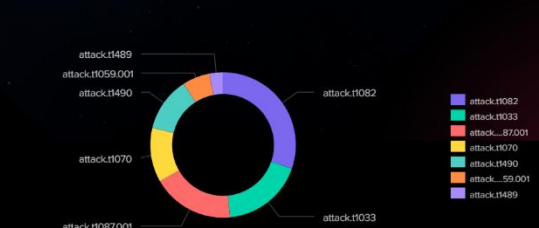
Detection Timeline



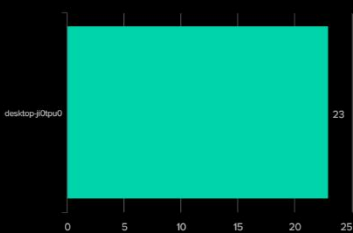
Top Detection Categories



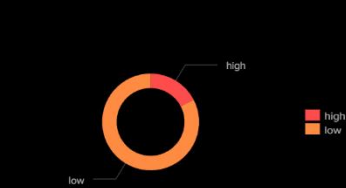
MITRE ATT&CK Techniques



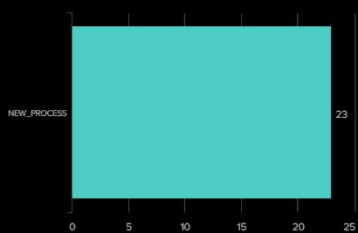
Top Affected Endpoints



Severity Breakdown



Detection by Event Type



Recent Detections

_time	Severi...	Detecti...	Endpoi...	Source...	External...	Event_Ty...	MIT...	Rule	Auth...
2026-07-28T15:37:19.000 +07:00	HIGH	Shadow Copies Deletion Using Operating Systems Utilities	desktop-j10tpu0	172.16.11.0	123.24.225.22.1	NEW_PROCESS	attack.t1070, attack.t1490	service.window_s_process_creation/proc_creation_win_susp_shadow_copies_deletion	_ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
2026-07-28T15:37:19.000 +07:00	LOW	Non Interactive PowerShell Process Spawned	desktop-j10tpu0	172.16.11.0	123.24.225.22.1	NEW_PROCESS	attack.t1059.00	service.window_s_process_creation/proc_creation_win_powershell_non_interactive_execution	_ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
2026-07-28T15:37:19.000	LOW	Suspicious Execution of	desktop-j10tpu0	172.16.11.0	123.24.225.22.1	NEW_PROCESS	attack.t108	service.window_s_process_crea	_ext-sigma-7a14fbc3-54d9-

< Prev 1 2 Next >

Suspicious Process Executions

_time	Endpoi...	User	Proce...	Command...	ParentPro...	PID	PPID	Hash	Detecti...
2026-07-28T15:37:19.000 +07:00	desktop-j10tpu0	DESKTOP-J10TPU0\longday	C:\Windows\system32\wssadmin.exe	"C:\Windows\system32\wssadmin.exe" resize shadowstorage /For=C: /On=C: /MaxSize=20%	C:\Windows\System32\Windows PowerShell\PowerShell\powershell.exe	8140	628	8cfabcc2196e4d096b7d155837c5f699ad7f55edb84571e4f8e03500b7a8b0	Shadow Copies Deletion Using Operating Systems Utilities
2026-07-28T15:37:19.000 +07:00	desktop-j10tpu0	DESKTOP-J10TPU0\longday	C:\Windows\System32\Windows PowerShell\powershell.exe	"powershell.exe" & [ysadmi resize shadowstorage /For=C: /On=C: /MaxSize=20%]	C:\Windows\System32\Windows PowerShell\powershell.exe	628	605	9785001b0dc7d55eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3	Non Interactive PowerShell Process Spawned
2026-07-28T15:37:19.000	desktop-j10tpu0	DESKTOP-J10TPU0\longday	C:\Windows\system32\HOSTNAME	"C:\Windows\system32\HOSTNAME"	C:\Windows\System32\Windows	7924	628	a90c3fb350a1fc6f6a6efa96079	Suspicious Execution of

< Prev 1 2 Next >

```

15:37:16 Shadow Copies Deletion Using Operating Systems Utilities → desktop-j10tpu0 ["event":{"COMMAND_LINE":"C:\\Windows\\system32\\vssadmin.exe"} resize shadowstorage /For=C: /On=C: /MaxSize=20%","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\
15:37:16 Suspicious Execution of Hostname → desktop-j10tpu0 ["event":{"COMMAND_LINE":"C:\\Windows\\system32\\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\HOSTNAME.EXE","HASH":"a90c3fb350a11c6f6a6efa9607987d92
15:37:16 Suspicious Execution of Hostname → desktop-j10tpu0 ["event":{"COMMAND_LINE":"C:\\Windows\\system32\\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\HOSTNAME.EXE","HASH":"a90c3fb350a11c6f6a6efa9607987d92
15:37:15 Non Interactive PowerShell Process Spawned → desktop-j10tpu0 ["event":{"BASE_ADDRESS":"14870197203456","COMMAND_LINE":"powershell.exe" & {vssadmin resize shadowstorage /For=C: /On=C: /MaxSize=20%},"FILE_IS_SIGNED":1,"FILE_PA
15:37:14 Shadow Copies Deletion Using Operating Systems Utilities → desktop-j10tpu0 ["event":{"BASE_ADDRESS":"148701871046656","COMMAND_LINE":"wbadmin delete catalog -quiet","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\wbadmin.e
15:37:13 Shadow Copies Deletion Using Operating Systems Utilities → desktop-j10tpu0 ["event":{"COMMAND_LINE":"wmic.exe shadowcopy delete","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\Wbem\\WMIC.exe","HASH":"da083fa720ab43aa6d4
15:37:12 Shadow Copies Deletion Using Operating Systems Utilities → desktop-j10tpu0 ["event":{"COMMAND_LINE":"vssadmin.exe delete shadows /all /quiet","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\vssadmin.exe","HASH":"8c1fabcc
15:37:09 Local Accounts Discovery → desktop-j10tpu0 ["event":{"COMMAND_LINE":"C:\\Windows\\system32\\whoami.exe","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\whoami.exe","HASH":"1d4092a04d99e8c8fe7085e63155952fee397449d38
15:37:09 Suspicious Execution of Hostname → desktop-j10tpu0 ["event":{"COMMAND_LINE":"C:\\Windows\\system32\\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\HOSTNAME.EXE","HASH":"a90c3fb350a11c6f6a6efa9607987d92
15:36:32 Stop Windows Service Win Sc.Exe → desktop-j10tpu0 ["event":{"COMMAND_LINE":"sc.exe stop spooler","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\sc.exe","HASH":"4d4092a04d99e8c8fe7085e63155952fee397449d38
15:36:31 Suspicious Execution of Hostname → desktop-j10tpu0 ["event":{"COMMAND_LINE":"C:\\Windows\\system32\\HOSTNAME.EXE","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\HOSTNAME.EXE","HASH":"a90c3fb350a11c6f6a6efa9607987d92
15:36:31 Local Accounts Discovery → desktop-j10tpu0 ["event":{"COMMAND_LINE":"C:\\Windows\\system32\\whoami.exe","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Windows\\system32\\whoami.exe","HASH":"1d4092a04d99e8c8fe7085e63155952fee397449d38

```

5. Analyst Technical Timeline

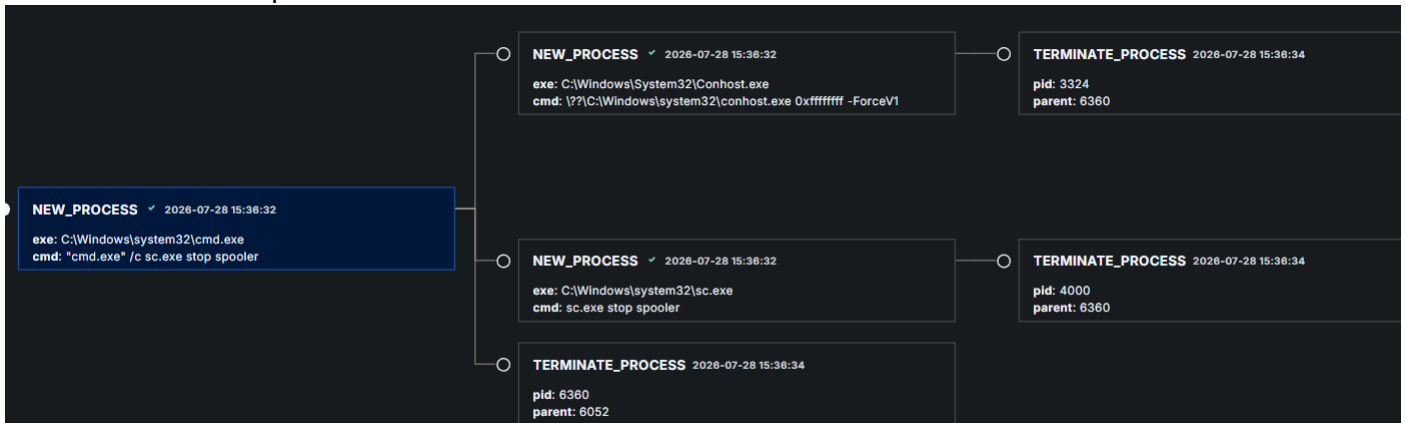
Time	Atomic Test	Command	ATT&CK Sub-technique
15:36:32	T1489-1	sc.exe stop spooler (qua cmd.exe /c)	T1489 - Service Stop
15:36:33	T1489-3	taskkill.exe /f /im spoolsv.exe (qua cmd.exe /c)	T1489 - Kill process khi service không stop clean
15:37:12	T1490-1	vssadmin.exe delete shadows /all /quiet (qua cmd.exe /c)	T1490.001-style - VSS deletion
15:37:13	T1490-2	wmic.exe shadowcopy delete (qua cmd.exe /c)	T1490 - VSS deletion qua WMI
15:37:14	T1490-3	wbadmin delete catalog -quiet (qua cmd.exe /c)	T1490 - Xóa Windows Backup catalog
15:37:14-15	T1490-9	"cmd.exe" /c reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows NT\SystemRestore" /v "DisableConfig" /t "REG_DWORD" /d "1" /f Khoá GPO policy — user không thể vào UI bật/tắt System Restore & reg add "HKLM\SOFTWARE\Policies\Microsoft\Windows NT\SystemRestore" /v "DisableSR" /t "REG_DWORD" /d "1" /f Ở nhánh Policy — tắt hẳn chức năng System Restore (không chỉ ẩn UI) & reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\SystemRestore" /v "DisableConfig" /t "REG_DWORD" /d "1" /f Lập lại tương tự nhưng ở nhánh CurrentVersion (không qua GPO) & reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\SystemRestore" /v "DisableSR" /t "REG_DWORD" /d "1" /f Tắt chức năng System Restore ở nhánh CurrentVersion	T1490 - Disable System Restore qua Registry
15:37:15-16	T1490-10	powershell.exe & {vssadmin resize shadowstorage /For=C: /On=C: /MaxSize=20%} (qua powershell.exe)	T1490 - Evasion variant (indirect purge)

15:36:31	NEW_PROCESS	Process (PID) whoami.exe (9072) Path: C:\Windows\system32\whoami.exe Command: "C:\Windows\system32\whoami.exe" User: ["COMMAND_LINE":"^\\?C:\\Windows\\system32\\whoami.exe";"
15:36:31	NEW_PROCESS	Process (PID) HOSTNAME.exe (8468) Path: C:\Windows\system32\HOSTNAME.exe Command: "C:\Windows\system32\HOSTNAME.exe" User: ["COMMAND_LINE":"^\\?C:\\Windows\\system32\\HOSTNAME.exe";"
15:36:31	DNS_REQUEST	Domain: api.smn.com ["DNS_TYPE":28,"DOMAIN_NAME":"api.smn.com","ERROR":0,"IP_ADDRESSES":["type: 5 api-smn-com-onsecrvice-wd-default-trafficmanager.net","type: 5 api-smn-com-ex-0003.akx-medge.net","typ
15:36:31	NEW_PROCESS	Process (PID) HtTxs.exe (5800) Path: C:\Program Files\WindowsApps\microsoft.windowscommunicationsapp_16095_14326_23242_0_x-ww_bakysj9dbbwa.HttTx.exe Command: "C:\Program Files\WindowsApps\microsof
15:36:32	NEW_PROCESS	Process (PID) svchost.exe (1140) Path: C:\Windows\System32\svchost.exe Command: C:\Windows\System32\svchost.exe -k wscntpr -c c:\ipSVC User: NT AUTHORITY\SYSTEM ["COMMAND_LINE":"C:\\Windows\\Systeme
15:36:32	NEW_PROCESS	Process (PID) cmd.exe (6360) Path: C:\Windows\system32\cmd.exe Command: "cmd.exe" /o so.exe stop spooler User: ["COMMAND_LINE":"","cmd.exe"/o so.exe stop spooler","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Win
15:36:32	NEW_PROCESS	Process (PID) Conhost.exe (3324) Path: C:\Windows\System32\Conhost.exe Command: \\??C:\Windows\system32\conhost.exe 0xffffffff -ForceV1 User: ["COMMAND_LINE":"","\\??C:\\Windows\\system32\\conhost.e
15:36:32	NEW_PROCESS	Process (PID) so.exe (4090) Path: C:\Windows\system32\so.exe Command: so.exe stop spooler User: DESKTOP-JIOTPU0longday ["COMMAND_LINE":"so.exe stop spooler","FILE_IS_SIGNED":1,"FILE_PATH":"C:\\Win
15:36:32	TERMINATE_PROCESS	PID: 2252 ["PARENT_PROCESS_ID":656,"PROCESS_ID":2252]
15:36:32	TERMINATE_PROCESS	PID: 9072 ["PARENT_PROCESS_ID":6052,"PROCESS_ID":9072]
15:36:32	TERMINATE_PROCESS	PID: 5468 ["PARENT_PROCESS_ID":6052,"PROCESS_ID":5468]
15:36:32	DNS_REQUEST	Domain: upad ["DNS_TYPE":1,"DOMAIN_NAME":"upad","ERROR":123,"PROCESS_ID":7644,"THREAD_ID":6624]
15:36:33	NEW_PROCESS	Process (PID) DllHost.exe (3484) Path: C:\Windows\system32DllHost.exe Command: C:\Windows\system32DllHost.exe /Processid:[AB8902B4-09CA-4BB6-B7BD-AF5979A0D0] User: DESKTOP-JIOTPU0longday ["BAS
15:36:33	NEW_PROCESS	Process (PID) cmd.exe (4698) Path: C:\Windows\system32\cmd.exe Command: "cmd.exe" /o taskkill.exe /f /im spoolv.exe ["cmd.exe"/o taskkill.exe /f /im spoolv.exe","FILE_IS_
15:36:33	NEW_PROCESS	Process (PID) Conhost.exe (4780) Path: C:\Windows\System32\Conhost.exe Command: \\??C:\Windows\system32\conhost.exe 0xffffffff -ForceV1 User: ["COMMAND_LINE":"","\\??C:\\Windows\\system32\\conhost.e
15:36:33	NEW_PROCESS	Process (PID) taskkill.exe (360) Path: C:\Windows\system32\taskkill.exe Command: taskkill.exe /f /im spoolv.exe User: DESKTOP-JIOTPU0longday ["COMMAND_LINE":"taskkill.exe /f /im spoolv.exe","FILE_I
15:36:33	NEW_PROCESS	Process (PID) RuntimeBroker.exe (3452) Path: C:\Windows\System32\RuntimeBroker.exe Command: C:\Windows\System32\RuntimeBroker.exe -Embedding User: DESKTOP-JIOTPU0longday ["BASE_ADDRESS":1460975461
15:36:33	SENSITIVE_PROCESS	["EVENTS":{"event":{"BASE_ADDRESS":1460975461,"COMMAND_LINE":"","C:\Windows\System32\RuntimeBroker.exe -Embedding User: DESKTOP-JIOTPU0longday ["BASE_ADDRESS":1460975461
15:36:34	CODE_IDENTITY	Hash: 56f8c2c1790c389373b4c3f5dc1977cf9f6ef0e0810ec11f9fe47f05e Path: C:\Windows\system32\taskkill.exe ["ACCESS_TIME":178522508342,"ATTRIBUTES":0,"CREATION_TIME":1575769791670,"ERROR":0,"FILE_I
15:36:34	TERMINATE_PROCESS	PID: 5880 ["PARENT_PROCESS_ID":776,"PROCESS_ID":5880]
15:36:34	TERMINATE_PROCESS	PID: 349 {"PARENT_PROCESS_ID":4608,"PROCESS_ID":349}
15:36:34	TERMINATE_PROCESS	PID: 4698 {"PARENT_PROCESS_ID":6052,"PROCESS_ID":4698}
15:36:34	TERMINATE_PROCESS	PID: 324 {"PARENT_PROCESS_ID":6360,"PROCESS_ID":324}
15:36:34	TERMINATE_PROCESS	PID: 1750 {"PARENT_PROCESS_ID":4698,"PROCESS_ID":1750}
15:36:34	TERMINATE_PROCESS	PID: 6360 {"PARENT_PROCESS_ID":6052,"PROCESS_ID":6360}
15:36:34	TERMINATE_PROCESS	PID: 4090 {"PARENT_PROCESS_ID":6360,"PROCESS_ID":4090}
15:36:34	DNS_REQUEST	Domain: 233.184.33.23.in-addr.arpa ["DNS_TYPE":12,"DOMAIN_NAME":"233.184.33.23.in-addr.arpa","ERROR":0,"IP_ADDRESSES":["type: 12 423-33-184-233.deploy.static.akamaitechnologies.co"],"PROCESS_ID":3120,
15:36:38	TERMINATE_PROCESS	PID: 3404 {"PARENT_PROCESS_ID":776,"PROCESS_ID":3404}
15:36:45	NEW_PROCESS	Process (PID) splunk-MonitorNoHndle.exe (7492) Path: C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHndle.exe Command: "C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNo
15:36:46	TERMINATE_PROCESS	PID: 7492 {"PARENT_PROCESS_ID":724,"PROCESS_ID":7492}
15:36:46	NEW_PROCESS	Process (PID) splunk-admon.exe (3252) Path: C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe Command: "C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe" User: ["COMMA
15:36:47	NEW_PROCESS	Process (PID) splunk-netmon.exe (4992) Path: C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe Command: "C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe" User: ["COMMA
15:36:47	DNS_REQUEST	Domain: dns.google IP Address: 8.8.4.4 ["DNS_TYPE":1,"DOMAIN_NAME":"dns.google","IP_ADDRESS":"8.8.4.4","MESSAGE_ID":4049,"PROCESS_ID":7644]
15:36:47	DNS_REQUEST	Domain: dns.google IP Address: 8.8.8.8 ["DNS_TYPE":1,"DOMAIN_NAME":"dns.google","IP_ADDRESS":"8.8.8.8","MESSAGE_ID":4049,"PROCESS_ID":7644]
15:36:48	DNS_REQUEST	Domain: prod-atm-wds-edge.trafficmanager.net CNAME: prod-agic-jw-6.japanwest.cloudapp.azure.com ["CNAME":"prod-agic-jw-6.southeastasia.cloudapp.azure.com","DNS_TYPE":5,"DOMAIN_NAME":"prod-atm-wds-e
15:36:48	DNS_REQUEST	Domain: nav-edge.smartscreen.microsoft.com CNAME: prod-atm-wds-edge.trafficmanager.net ["CNAME":"prod-atm-wds-edge.trafficmanager.net","DNS_TYPE":5,"DOMAIN_NAME":"nav-edge.smartscreen.microsoft.com","MI
15:36:48	DNS_REQUEST	Domain: nav-edge.smartscreen.microsoft.com CNAME: prod-atm-wds-edge.trafficmanager.net ["CNAME":"prod-atm-wds-edge.trafficmanager.net","DNS_TYPE":5,"DOMAIN_NAME":"nav-edge.smartscreen.microsoft.com","MI
15:36:48	DNS_REQUEST	Domain: prod-atm-wds-edge.trafficmanager.net CNAME: prod-agic-jw-6.japanwest.cloudapp.azure.com ["CNAME":"prod-agic-jw-6.japanwest.cloudapp.azure.com","DNS_TYPE":5,"DOMAIN_NAME":"prod-atm-wds-edge.traf
15:36:48	DNS_REQUEST	Domain: prod-agic-jw-6.japanwest.cloudapp.azure.com IP Address: 135.149.147.5 ["DNS_TYPE":1,"DOMAIN_NAME":"prod-agic-jw-6.japanwest.cloudapp.azure.com","IP_ADDRESS":"135.149.147.5","MESSAGE_ID":42711,"
15:36:48	TERMINATE_PROCESS	PID: 3252 {"PARENT_PROCESS_ID":724,"PROCESS_ID":3252}
15:36:48	TERMINATE_PROCESS	PID: 4992 {"PARENT_PROCESS_ID":724,"PROCESS_ID":4992}
15:36:49	NEW_PROCESS	Process (PID) splunk-powershell.exe (3132) Path: C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe Command: "C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" Us
15:36:50	NEW_PROCESS	Process (PID) splunk-powershell.exe (4600) Path: C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe Command: "C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" -ps
15:36:50	TERMINATE_PROCESS	PID: 3132 {"PARENT_PROCESS_ID":724,"PROCESS_ID":3132}
15:36:51	DNS_REQUEST	Domain: 97.150.171.150.in-addr.arpa ["DNS_TYPE":12,"DOMAIN_NAME":"97.150.171.150.in-addr.arpa","ERROR":9003,"PROCESS_ID":3120,"THREAD_ID":5060]
15:36:51	NETWORK_CONNECTIONS	Connections: 2 PID: 4 {"NETWORK_ACTIVITY":{"DESTINATION":{"IP_ADDRESS":"172.16.1.255","PORT":137},"IS_OUTGOING":1,"PROTOCOL":"udp","SOURCE":{"IP_ADDRESS":"172.16.1.16","PORT":137},"TIME
15:36:51	NETWORK_CONNECTIONS	Connections: 1 Process (PID): backgroundTaskHost.exe (7744) ["COMMAND_LINE":"^\\?C:\\Windows\\system32\\backgroundTaskHost.exe";"-ServerName:ShellFeedsUI.AppXnJ6SkZd1stnzzt2zng5ctmk3z7pn.mca","FILE_I
15:36:51	NETWORK_CONNECTIONS	Connections: 41 Process (PID): mcsedgewebview2.exe (7640) ["COMMAND_LINE":"^\\?C:\\Program Files\\x86\\Microsoft\\EdgeWebView\\Application\\Shl6.0.4078.99\\mcsedgewebview2.exe";"-utility-shell","UTILITY_S
15:36:51	NEW_PROCESS	Process (PID) splunk-region.exe (3454) Path: C:\Program Files\SplunkUniversalForwarder\bin\splunk-region.exe Command: "C:\Program Files\SplunkUniversalForwarder\bin\splunk-region.exe" User: ["COMMA

[illegible]

10-37-18	TERMINATE_PROCESS	PID: 17924	["PARENT_PROCESS_ID":6289,"PROCESS_ID":17924]	
10-37-18	TERMINATE_PROCESS	PID: 7592	["PARENT_PROCESS_ID":6289,"PROCESS_ID":7592]	
10-37-18	TERMINATE_PROCESS	PID: 7580	["PARENT_PROCESS_ID":776,"PROCESS_ID":7580]	
10-37-18	TERMINATE_PROCESS	PID: 8140	["PARENT_PROCESS_ID":6289,"PROCESS_ID":8140]	
10-37-19	CODE_IDENTITY	Hash: 45c44b43f9e27b1f83172b60801e8bc11f1b9708513ea343a78744b6f29259	Path: 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T1489 – Service Stop



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07-28 15:36:32

Q Quick Search

ADD FILTER

07-28 15:36:52	TERMINATE_PROCESS	PID: 5688	!{"PARENT_PROCESS_ID":724,"PROCESS_ID":5688}
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07-28 15:36:08	✓ NEW_PROCESS	Process (PID): TiWorker.exe (4364)	Path: C:\Windows\winxsl\amd64_microsoft
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07-28 15:36:28	✓ NEW_PROCESS	Process (PID): svchost.exe (3640)	Path: C:\Windows\system32\svchost.exe
07-28 15:36:28	✓ NEW_PROCESS	Process (PID): taskhostw.exe (6000)	Path: C:\Windows\system32\taskhostw.exe
07-28 15:36:29	✓ NEW_PROCESS	Process (PID): backgroundTaskHost.exe (7740)	Path: C:\Windows\system32\backg
07-28 15:36:29	THREAD_INJECTION	PID: 440	---> 5020 !{"EVENTS":[{"event":{"ACCESS_FLAGS":2097151,"PARENT_PRO
07-28 15:36:29	DNS_REQUEST	Domain: 71.44.207.4.in-addr.arpa. !{"DNS_TYPE":12,"DOMAIN_NAME":"71.44.207.4	
07-28 15:36:30	TERMINATE_PROCESS	PID: 6852	!{"PARENT_PROCESS_ID":5020,"PROCESS_ID":6852}
07-28 15:36:30	TERMINATE_PROCESS	PID: 6000	!{"PARENT_PROCESS_ID":1232,"PROCESS_ID":6000}
07-28 15:36:30	TERMINATE_PROCESS	PID: 5000	!{"PARENT_PROCESS_ID":1232,"PROCESS_ID":5000}

70728f203d2d7b67d9ee7ea6a686a10

SCROLL INTO VIEW

GOTO PARENT

VIEW TRACELINE

COPY EVENT

BUILD DAR FILE

TIME

2028-07-28 15:36:32

HOSTNAME

desktop-jj0tuo

EVENT ROUTING

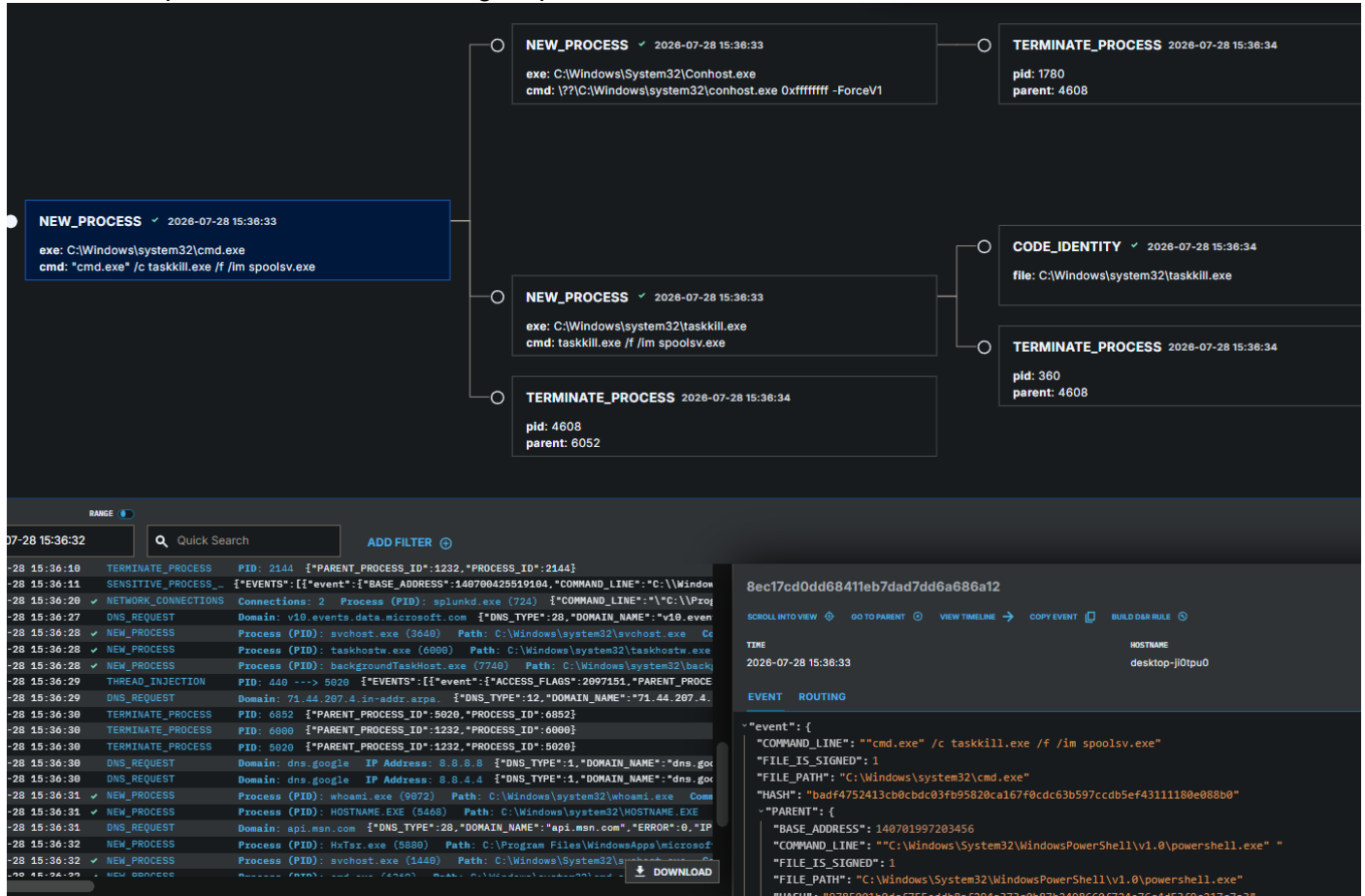
```

- "event": {
  "COMMAND_LINE": "cmd.exe /c sc.exe system spooler"
  "FILE_IS_SIGNED": 1
  "FILE_PATH": "C:\Windows\system32\cmd.exe"
  "HASH": "badf4752413cb0cdbc03f695820a167f0cdc63b597ccdb5ef4311180e088b0"
  "PARENT": {
    "BASE_ADDRESS": 140701997203456
    "COMMAND_LINE": "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
    "FILE_IS_SIGNED": 1
    "FILE_PATH": "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
  }
}

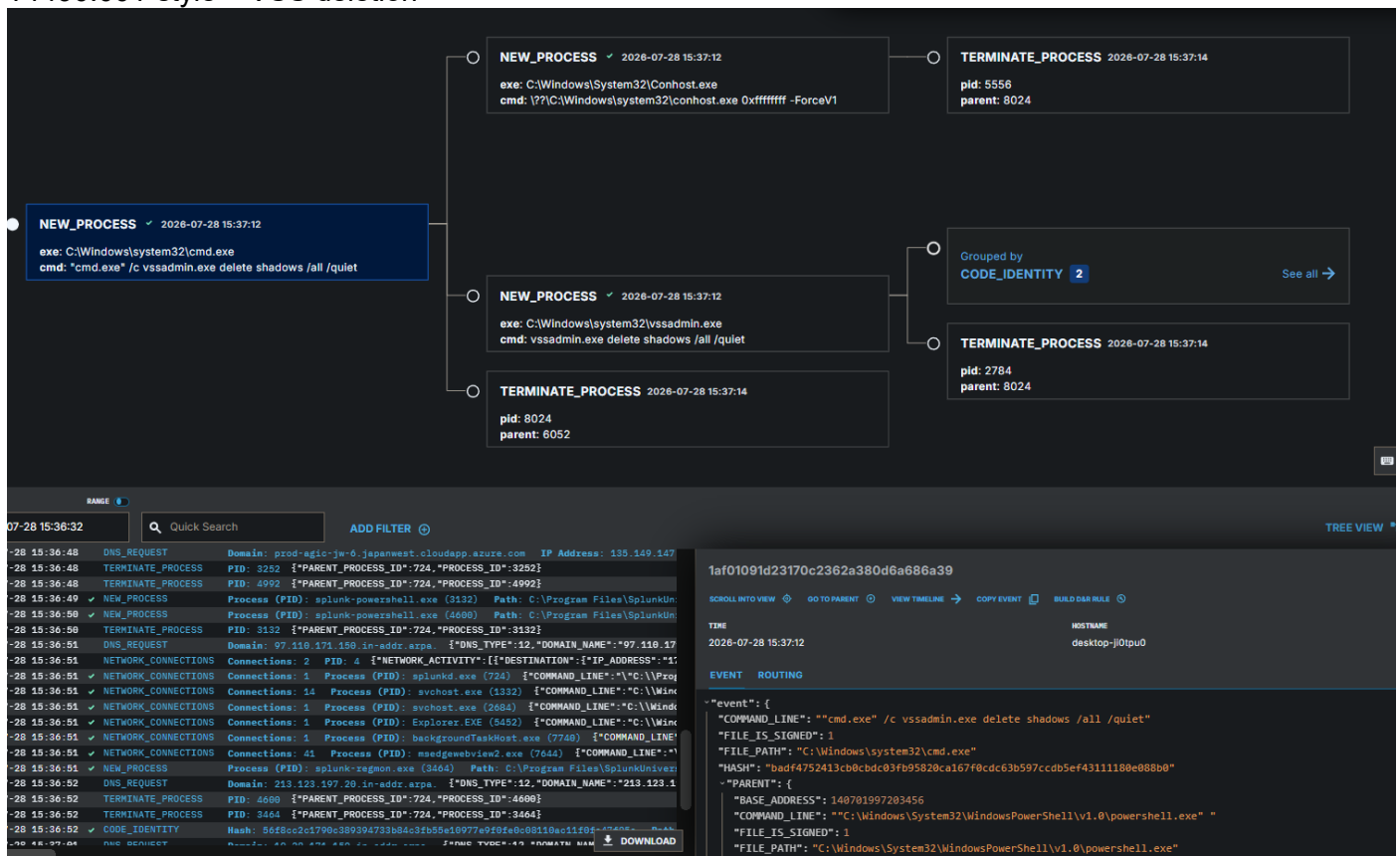
```

DOWNLOAD

T1489 – Kill process khi service không stop clean



T1490.001-style – VSS deletion



T1490 – VSS deletion qua WMI

T1490 – Disable System Restore qua Registry

```
~"event": {
  "COMMAND_LINE": "\\?\\C:\\Windows\\system32\\conhost.exe 0xffffffff -ForceV1"
  "FILE_IS_SIGNED": 1
  "FILE_PATH": "C:\\Windows\\System32\\Conhost.exe"
  "HASH": "b02ee54fb2ec69673386d41119ee8ed083a6eab3bfca6aa2155d20ce68ef8963"
  ~"PARENT": {
    "COMMAND_LINE":
      ""cmd.exe" /c reg add "HKLM\\SOFTWARE\\Policies\\Microsoft\\Windows NT\\SystemRestore" /v
      "DisableConfig" /t "REG_DWORD" /d "1" /f & reg add "HKLM\\SOFTWARE\\Policies\\Microsoft\\Windows
      NT\\SystemRestore" /v "DisableSR" /t "REG_DWORD" /d "1" /f & reg add
      "HKLM\\SOFTWARE\\Microsoft\\Windows NT\\CurrentVersion\\SystemRestore" /v "DisableConfig" /t
      "REG_DWORD" /d "1" /f & reg add "HKLM\\SOFTWARE\\Microsoft\\Windows
      NT\\CurrentVersion\\SystemRestore" /v "DisableSR" /t "REG_DWORD" /d "1" /f
    "FILE_IS_SIGNED": 1
    "FILE_PATH": "C:\\Windows\\system32\\cmd.exe"
    "HASH": "badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef4311180e088b0"
    "PARENT_ATOM": "ed5898db1a7a419e4e1422286a6865e7"
    "PARENT_PROCESS_ID": 6052
    "PROCESS_ID": 7728
    "THIS_ATOM": "d0c429fed8f0fa15d30b515a6a686a3b"
    "TIMESTAMP": 1785227835214
  }
  "PARENT_PROCESS_ID": 7728
  "PROCESS_ID": 8988
}
```

T1490 – Evasion variant (indirect purge)

```
event": {
  "BASE_ADDRESS": 140701997203456
  "COMMAND_LINE": "powershell.exe" & {vssadmin resize shadowstorage /For=C: /On=C: /MaxSize=20%}
  "FILE_IS_SIGNED": 1
  "FILE_PATH": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
  "HASH": "9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3"
  "MEMORY_USAGE": 95846400
  ~"PARENT": {
    "BASE_ADDRESS": 140701997203456
    "COMMAND_LINE": ""C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
    "FILE_IS_SIGNED": 1
    "FILE_PATH": "C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe"
    "HASH": "9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3"
    "MEMORY_USAGE": 76316672
    "PARENT_ATOM": "a44ff5ff0e5e99c28fddaec06a6861d8"
    "PARENT_PROCESS_ID": 8784
    "PROCESS_ID": 6052
    "THIS_ATOM": "ed5898db1a7a419e4e1422286a6865e7"
    "THREADS": 20
    "TIMESTAMP": 1785226726270
    "USER_NAME": "DESKTOP-3I0TPU0\\longday"
  }
  "PARENT_PROCESS_ID": 6052
  "PROCESS_ID": 6280
  "THREADS": 18
  "USER_NAME": "DESKTOP-3I0TPU0\\longday"
}
```

6. Endpoint and SIEM Evidence

LimaCharlie detection evidence

Các detection liên quan trực tiếp:

- Stop Windows Service Via Sc.EXE
- Shadow Copies Deletion Using Operating Systems Utilities
- Non Interactive PowerShell Process Spawned
- Detection Shadow Copies Deletion Using Operating Systems Utilities xuất hiện cho:
 - vssadmin.exe delete shadows
 - WMIC.exe shadowcopy delete
 - wbadm.in.exe delete catalog
 - vssadmin.exe resize shadowstorage

Splunk evidence

Splunk dashboard ghi nhận trong cửa sổ quan sát:

- 23 detection.
- Một endpoint bị ảnh hưởng.
- Bốn High/Critical alert.
- Bốn detection thuộc nhóm Shadow Copies.
- Một detection thuộc nhóm Stop Windows Service.
- attack.t1489 và attack.t1490 xuất hiện trong MITRE mapping.
- Các detection mới nhất được hiển thị tại khoảng 15:37:19.

Số lượng 23 detection bao gồm cả các sự kiện hỗ trợ như hostname.exe, whoami.exe và PowerShell wrapper. Đây không phải 23 hành vi Impact độc lập.

```
7/28/26 3:36:35.000 PM { [-]
  author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
  cat: Stop Windows Service Via Sc.EXE
  detect: { [-]
    event: { [-]
      COMMAND_LINE: sc.exe stop spooler
      FILE_IS_SIGNED: 1
      FILE_PATH: C:\Windows\system32\sc.exe
      HASH: 41f067c3a11b02fe39947f9eba68ae5c7cb5bd1872a6009a4cd1506554a9aba9
      MEMORY_USAGE: 36864
      PARENT: { [-]
        COMMAND_LINE: "cmd.exe" /c sc.exe stop spooler
        FILE_IS_SIGNED: 1
        FILE_PATH: C:\Windows\system32\cmd.exe
        HASH: badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef43111180e088b0
        PARENT_ATOM: ed5898db1a7a419e4e1422286a6865e7
        PARENT_PROCESS_ID: 6052
        PROCESS_ID: 6360
        THIS_ATOM: 770728f203d2d7b67d9ee7ea6a686a10
        TIMESTAMP: 1785227792323
      }
      PARENT_PROCESS_ID: 6360
      PROCESS_ID: 4000
      USER_NAME: DESKTOP-JI0TPU0\longday
    }
    routing: { [+]
  }
}
detect_id: 335c6e23-4c89-4dc6-906c-ee6d6a686a10
detect_mtd: { [+]
}
gen_time: 1785227792974
link: https://app.limacharlie.io/orgs/cbe5ce69-32fe-43bb-a298-2a274f788a40/sensors/16b750c9-3256-4ca1-b2c7-864
namespace: general
```

```
7/28/26 3:37:15.000 PM { [-]
  author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
  cat: Shadow Copies Deletion Using Operating Systems Utilities
  detect: { [-]
    event: { [-]
      COMMAND_LINE: vssadmin.exe delete shadows /all /quiet
      FILE_IS_SIGNED: 1
      FILE_PATH: C:\Windows\system32\vssadmin.exe
      HASH: 8c1fabcc2196e4d096b7d155837c5f699ad7f55edbf84571e4f8e03500b7a8b0
      MEMORY_USAGE: 36864
      PARENT: { [-]
        COMMAND_LINE: "cmd.exe" /c vssadmin.exe delete shadows /all /quiet
        FILE_IS_SIGNED: 1
        FILE_PATH: C:\Windows\system32\cmd.exe
        HASH: badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef43111180e088b0
        PARENT_ATOM: ed5898db1a7a419e4e1422286a6865e7
        PARENT_PROCESS_ID: 6052
        PROCESS_ID: 8024
        THIS_ATOM: 1af01091d23170c2362a380d6a686a39
        TIMESTAMP: 1785227832714
      }
      PARENT_PROCESS_ID: 8024
      PROCESS_ID: 2784
      USER_NAME: DESKTOP-JI0TPU0\longday
    }
    routing: { [+]
  }
}
detect_id: 93c0547a-9a21-4bac-81b9-b4036a686a39
detect_mtd: { [+]
}
gen_time: 1785227833542
link: https://app.limacharlie.io/orgs/cbe5ce69-32fe-43bb-a298-2a274f788a40/sensors/16b750c9-3256-4ca1-b2c7-864
namespace: general
```

```
7/28/26      { [-]
3:37:16.000 PM author: .ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Shadow Copies Deletion Using Operating Systems Utilities
detect: { [-]
  event: { [-]
    COMMAND_LINE: wmic.exe shadowcopy delete
    FILE_IS_SIGNED: 1
    FILE_PATH: C:\Windows\System32\Wbem\WMIC.exe
    HASH: da603fa720ab43aa6d4d36aa9fdb828ab9645523eabaac209af6451d9b4d757
    MEMORY_USAGE: 36864
    PARENT: { [-]
      BASE_ADDRESS: 140703100829696
      COMMAND_LINE: "cmd.exe" /c wmic.exe shadowcopy delete
      FILE_IS_SIGNED: 1
      FILE_PATH: C:\Windows\system32\cmd.exe
      HASH: badf4752413cb0bcd03fb95820ca167f0cdc63b597ccdb5ef4311180e088b0
      MEMORY_USAGE: 4575232
      PARENT_ATOM: ed5898db1a7a419e4e1422286a6865e7
      PARENT_PROCESS_ID: 6052
      PROCESS_ID: 2960
      THIS_ATOM: ea5960a6e5ddf2751bd32df56a686a39
      THREADS: 3
      TIMESTAMP: 1785227833136
      USER_NAME: DESKTOP-JI0TPU0\longday
    }
    PARENT_PROCESS_ID: 2960
    PROCESS_ID: 8300
    USER_NAME: DESKTOP-JI0TPU0\longday
  }
  routing: { [+]}
}
detect_id: 1c120dab-c9a8-4528-9a28-d7e26a686a39
detect_mtd: { [-]}
}
Sen.time: 1785227833629
link: https://app.limacharlie.io/orgs/cbe5ce69-32fe-43bb-a298-2a274f788a40/sensors/16b750c9-3256-4ca1-b2c
namespace: general
routing: { [+]}
}
```

```
7/28/26      { [-]
3:37:17.000 PM author: .ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Shadow Copies Deletion Using Operating Systems Utilities
detect: { [-]
  event: { [-]
    BASE_ADDRESS: 140701871046656
    COMMAND_LINE: wbadmin delete catalog -quiet
    FILE_IS_SIGNED: 1
    FILE_PATH: C:\Windows\system32\wbadmin.exe
    HASH: 94239bd6d7502b11243f47d7ea2b595843ccde58368c2c9f1b14226e6867f6
    MEMORY_USAGE: 7839744
    PARENT: { [-]
      BASE_ADDRESS: 140703100829696
      COMMAND_LINE: "cmd.exe" /c wbadmin delete catalog -quiet
      FILE_IS_SIGNED: 1
      FILE_PATH: C:\Windows\system32\cmd.exe
      HASH: badf4752413cb0bcd03fb95820ca167f0cdc63b597ccdb5ef4311180e088b0
      MEMORY_USAGE: 4444160
      PARENT_ATOM: ed5898db1a7a419e4e1422286a6865e7
      PARENT_PROCESS_ID: 6052
      PROCESS_ID: 4276
      THIS_ATOM: caff9e28475cdafa181e70d36a686a3a
      THREADS: 3
      TIMESTAMP: 1785227834573
      USER_NAME: DESKTOP-JI0TPU0\longday
    }
    PARENT_PROCESS_ID: 4276
    PROCESS_ID: 4004
    THREADS: 6
    USER_NAME: DESKTOP-JI0TPU0\longday
  }
}
```

```
7/28/26 3:37:19.000 PM { [-]
author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Non Interactive PowerShell Process Spawned
detect: { [-]
event: { [-]
BASE_ADDRESS: 140701997203456
COMMAND_LINE: "powershell.exe" & (vssadmin resize shadowstorage /For=C: /On=C: /MaxSize=20%)
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
HASH: 9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3
MEMORY_USAGE: 95846400
PARENT: { [-]
}
PARENT_PROCESS_ID: 6052
PROCESS_ID: 6280
THREADS: 10
USER_NAME: DESKTOP-JI0TPU0\longday
}
routing: { [+]
}
}
detect_id: 5497c46a-58a5-484c-bb7a-28976a686a3c
detect_mtd: { [-]
}
gen_time: 1785227836793
link: https://app.limacharlie.io/orgs/cbe5ce69-32fe-43bb-a298-2a274f788a40/sensors/16b750c9-3256-4ca1-b2c7-8640b3a9d745/timeline
namespace: general
routing: { [-]
}
}
```

```
7/28/26 3:37:19.000 PM { [-]
author: _ext-sigma-7a14fbc3-54d9-4b4d-8700-61eddada04f0[bulk][segment]
cat: Non Interactive PowerShell Process Spawned
detect: { [-]
event: { [-]
BASE_ADDRESS: 140701997203456
COMMAND_LINE: "powershell.exe" & (vssadmin resize shadowstorage /For=C: /On=C: /MaxSize=20%)
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
HASH: 9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3
MEMORY_USAGE: 95846400
PARENT: { [-]
BASE_ADDRESS: 140701997203456
COMMAND_LINE: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
FILE_IS_SIGNED: 1
FILE_PATH: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
HASH: 9785001b0dcf755eddb8af294a373c0b87b2498660f724e76c4d53f9c217c7a3
MEMORY_USAGE: 76316672
PARENT_ATOM: a44ff5ff0e5e99c28fddaec06a6861d8
PARENT_PROCESS_ID: 8784
PROCESS_ID: 6052
THIS_ATOM: ed5898db1a7a419e4e142286a6865e7
THREADS: 20
TIMESTAMP: 1785226726270
USER_NAME: DESKTOP-JI0TPU0\longday
}
PARENT_PROCESS_ID: 6052
PROCESS_ID: 6280
THREADS: 18
USER_NAME: DESKTOP-JI0TPU0\longday
}
routing: { [+]
}
}
detect_id: 5497c46a-58a5-484c-bb7a-28976a686a3c
detect_mtd: { [-]
}
gen_time: 1785227836793
link: https://app.limacharlie.io/orgs/cbe5ce69-32fe-43bb-a298-2a274f788a40/sensors/16b750c9-3256-4ca1-b2c7-8640b3a9d745/timeline
namespace: general
routing: { [-]
}
}
```

7. MITRE ATT&CK Mapping

Thành phần	Ảnh xạ
Tactic	Impact
Technique 1	T1489 - Service Stop
Technique 2	T1490 - Inhibit System Recovery
T1489 utilities	sc.exe, taskkill.exe
T1490 utilities	vssadmin.exe, WMIC.exe, wbadmin.exe, reg.exe, powershell.exe
Data sources	Process creation, command line, process relationship, Registry telemetry và service events
Detection basis	Native utility execution, high-risk arguments và chuỗi hành vi trên cùng endpoint

8. Impact and Risk Assessment

Chuỗi T1489 kết hợp T1490 thường xuất hiện trước các hành vi:

- Mã hóa dữ liệu.
- Xóa hoặc phá hủy dữ liệu.
- Vô hiệu hóa backup.
- Ngăn người dùng khôi phục hệ thống.
- Giải phóng file đang bị service khóa trước khi mã hóa.